Design B - Liveboard læses fra en summary tabel



Client

API

Liveboard

Commands (writes)

RenameRouteCommand

CancelTripCommand

RenameRouteHandler

CancelTripHandler

Queries (reads)

GetLiveboardQuery

GetLiveBoardHandler

Live-tavlen læses fra route_overview_summary
(ingen beregning ved forespørgsel)

Booking

Commands (writes)

MakeBookingCommand

MakeBookingHandler

Transaction (MakeBooking)

- Tjek request_id(idempotency)
- Lås tur (SELECT __ FOR UPDATE)
- Tjek trips.status = “Scheduled”
- Opret booking

PublishedPlan

Queries (reads)

GetPublishedPlanQuery

GetPublishedPlanHandler

Commands (writes)

PublishPlanCommand

PublishPlanHandler

Transaction (PublishPlan)

- Læs routes + trips (for vindue)
- Beregn værdier pr. rute
- Gem i published_plan_snapshot
- Commit

Idempotency: samme plan + vindue ellers → SnapshotConflictException



Database

routes

id	text	PK
short_name	text	

trips

id	text	PK
route_id	text	FK
scheduled_departure_utc	timestampz	
status	text	

bookings

request_id	text	PK
trip_id	text	FK
booked_at_utc	timestampz	

route_overwiev_summary

route_id	text	PK
window_start_utc	timestampz	PK
window_end_utz	timestampz	
route_name	text	
scheduled_trip_count	integer	
next_departure_utc	timestampz	

published_plan_snapshot

id	text	PK
window_start_utc	timestampz	
window_end_utc	timestampz	
route_id	text	
route_name	text	
scheduled_trip_count	integer	
next_departure_utc	timestampz	

Fordele
Meget hurtig at læse (ingen beregning ved forespørgsel).
Mindre belastning på databasen ved mange forespørgsler.
Velegnet hvis det samme vindue vises ofte.

Ulemper
Flere ting at holde opdateret (trips + summary).
Risiko for inkonsistens hvis en handler glemmer at opdatere summary-tabellen.
Understøtter primært et tidsvindue (eller kræver flere rækker pr. vindue).
Mere kompleksitet